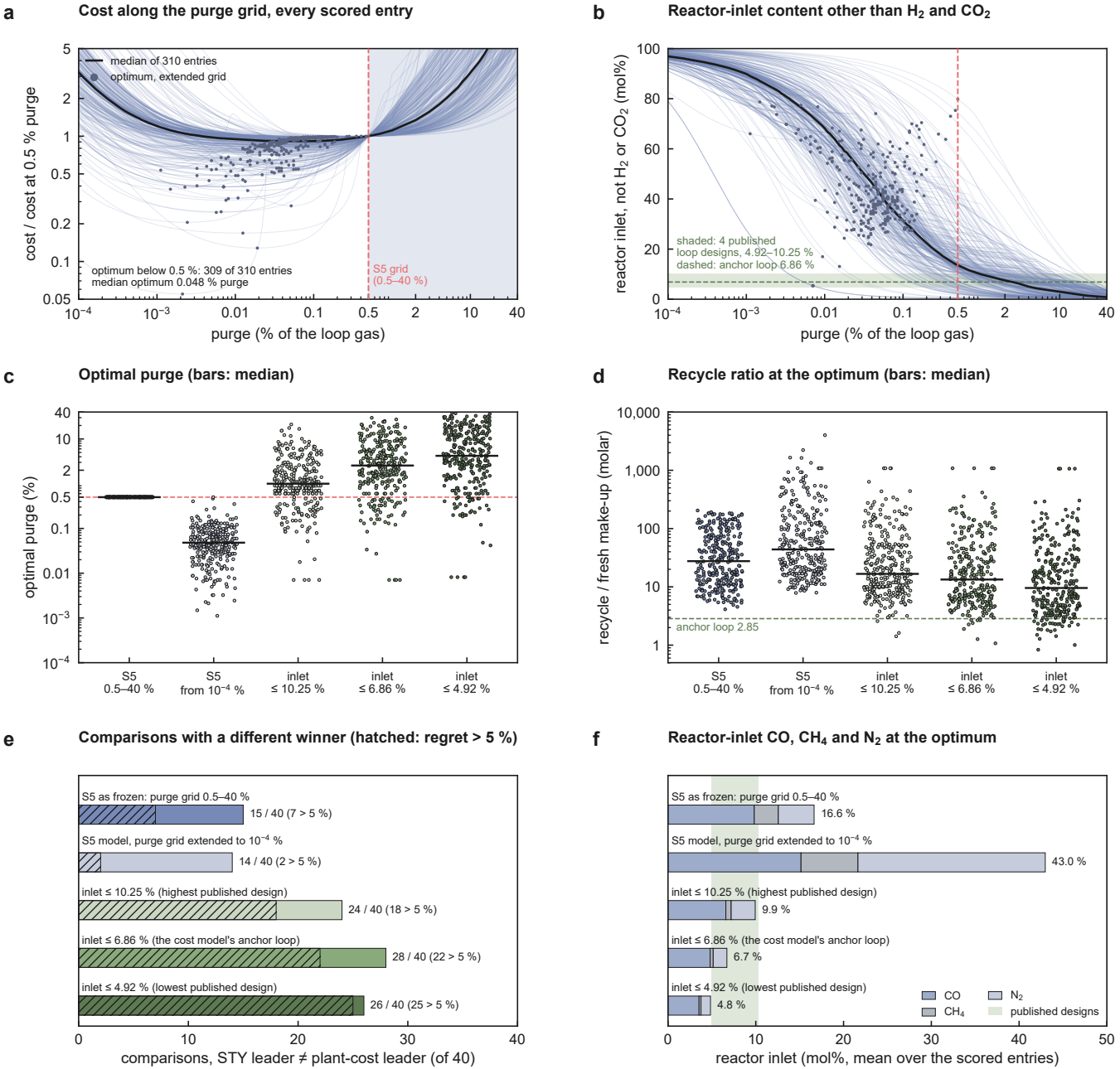

a
Cost along the purge grid, every scored entry
median of 310 entries
optimum, extended grid
optimum below 0.5 %: 309 of 310 entries
median optimum 0.048 % purge
S5 grid
(0.5–40 %)
cost / cost at 0.5 % purge
purge (% of the loop gas)
b
Reactor-inlet content other than H2 and CO2
shaded: 4 published
loop designs, 4.92–10.25 %
dashed: anchor loop 6.86 %
reactor inlet, not H2 or CO2 (mol%)
purge (% of the loop gas)
c
Optimal purge (bars: median)
optimal purge (%)
S5
0.5–40 %
S5
from 10−4 %
inlet
≤ 10.25 %
inlet
≤ 6.86 %
inlet
≤ 4.92 %
d
Recycle ratio at the optimum (bars: median)
recycle / fresh make-up (molar)
anchor loop 2.85
S5
0.5–40 %
S5
from 10−4 %
inlet
≤ 10.25 %
inlet
≤ 6.86 %
inlet
≤ 4.92 %
e
Comparisons with a different winner (hatched: regret > 5 %)
S5 as frozen: purge grid 0.5–40 %
15 / 40 (7 > 5 %)
S5 model, purge grid extended to 10−4 %
14 / 40 (2 > 5 %)
inlet ≤ 10.25 % (highest published design)
24 / 40 (18 > 5 %)
inlet ≤ 6.86 % (the cost model's anchor loop)
28 / 40 (22 > 5 %)
inlet ≤ 4.92 % (lowest published design)
26 / 40 (25 > 5 %)
comparisons, STY leader ≠ plant-cost leader (of 40)
f
Reactor-inlet CO, CH4 and N2 at the optimum
S5 as frozen: purge grid 0.5–40 %
16.6 %
S5 model, purge grid extended to 10−4 %
43.0 %
inlet ≤ 10.25 % (highest published design)
9.9 %
inlet ≤ 6.86 % (the cost model's anchor loop)
6.7 %
inlet ≤ 4.92 % (lowest published design)
4.8 %
CO
N2
CH4
published designs
reactor inlet (mol%, mean over the scored entries)